Probabilistic Multiple Prototypes Contrastive Alignment for Cross-Domain Semantic Segmentation of Remote Sensing Images

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***Abstract*—Unsupervised domain adaptation (UDA) aims to transfer knowledge from a labeled source domain to an unlabeled target domain to improve model generalization, and has been widely used in remote sensing image (RSI) semantic segmentation. However, complex intradomain category distributions and mismatched cross-domain category priors hinder accurate domain alignment in existing UDA methods. To address these challenges, we propose a probabilistic multiple prototypes contrastive alignment (PMPCA) UDA method. First, we develop a probabilistic multiple prototypes learning (PMPL) module, which constructs multiple prototypes along with probabilistic parameters to represent both the central tendency and dispersion of category distributions. This module enables fine-grained preservation of category semantics in both the source and target domains. Second, we introduce a prior-weighted contrastive (PWC) constraint, which establishes prototype-level contrastive learning between the source and target domains. It pulls prototypes toward their corresponding categories while pushing them away from other categories, thereby promoting structural alignment across domains. Furthermore, we estimate category frequencies in both domains and use them as prior weights for adaptive calibration, mitigating bias induced by category prior and refining alignment. Finally, we design an uncertainty-guided curriculum adaptation (UCA) strategy, which quantifies the difficulty of the target domain based on prediction uncertainty and enables progressive UDA from easy to hard cases. This strategy reduces the interference of noisy pseudo-labels. Extensive experiments on six UDA settings demonstrate that PMPCA consistently outperforms state-of-the-art methods and generalizes well across spatial regions and spectral bands.**

***Index Terms*—Contrastive learning, curriculum learning, probabilistic prototype, semantic segmentation, unsupervised domain adaptation (UDA).**

I. Introduction

REMOTE sensing image (RSI) semantic segmentation is a fundamental task in Earth observation that assigns a category label (e.g., building, water, and road) to each instance pixel for detailed land-surface understanding [1]. However, RSIs acquired from different domains often exhibit significant discrepancies due to dynamic real-world factors, such as sensor types and weather conditions, leading to shifts between the source and target domains [2]. As a result, models that perform well on the source domain frequently suffer substantial performance degradation when directly applied to the target domain [3].

To mitigate this issue, unsupervised domain adaptation (UDA) techniques have been extensively studied for RSI semantic segmentation [3]. The goal of UDA is to reduce discrepancies between the source and target domains, thereby enabling effective knowledge transfer without target domain annotations [3]. Existing UDA methods for RSI semantic segmentation can be broadly grouped into two types [4], [5]. Adversarial training (AT)-based methods introduce an external discriminator to establish an adversarial game with the segmentation model. Through this process, the model learns to confuse the discriminator, making it unable to distinguish whether the input comes from the source or target domain, thereby enabling direct domain alignment [6]. This process can be conducted at multiple levels, including the input, intermediate, and output levels [1], [7], [8]. These levels correspond to reducing appearance, feature, and prediction discrepancies across domains, respectively.

In contrast, ST-based methods rely on an internal closed-loop collaboration between the teacher and student models, enabling progressive domain alignment [36]. The teacher model generates pseudo-labels for the target domain, while the student model is jointly supervised by true labels from the source domain and pseudo-labels from the target domain. The teacher model parameters are updated as an exponential moving average (EMA) of the student model parameters, whereas the student model parameters are optimized via gradient backpropagation [10]. As ST proceeds, the quality of pseudo-labels typically improves, thereby reducing discrepancies across domains. To alleviate pseudo-label noise, various enhancement strategies have been proposed, including fixed and dynamic confidence-threshold filtering [11] and uncertainty-aware weighting [12].

Prototypes are commonly constructed by averaging the features of instances within the same category, thereby capturing the common patterns of that category [13], [14], [15]. This allows instances to be clustered based on their feature distances to the prototypes and provides an unsupervised representation of category-level semantics, which significantly strengthens existing UDA methods. Specifically, in AT-based methods, prototypes facilitate accurate category alignment between the source and target domains, alleviating semantic

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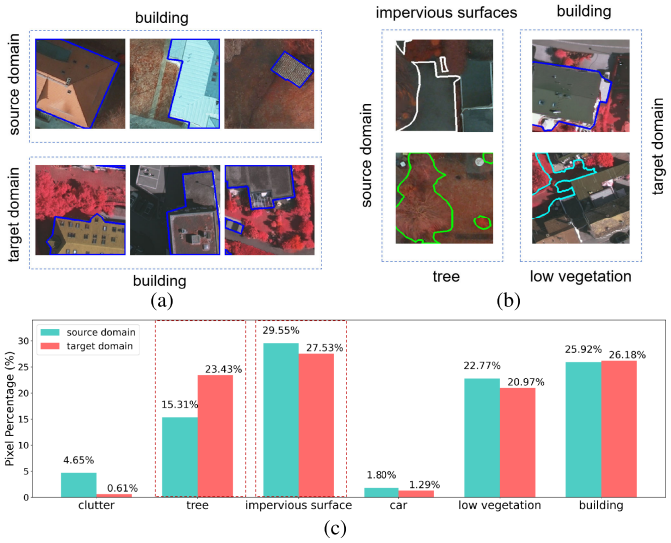


Fig. 1. Schematic illustration of the challenges of UDA for RSI semantic segmentation. The source domain is the Potsdam dataset, and the target domain is the Vaihingen dataset. (a) Intracategory variability within domains. (b) Intercategory similarity across domains. (c) Category-frequency mismatch across domains.

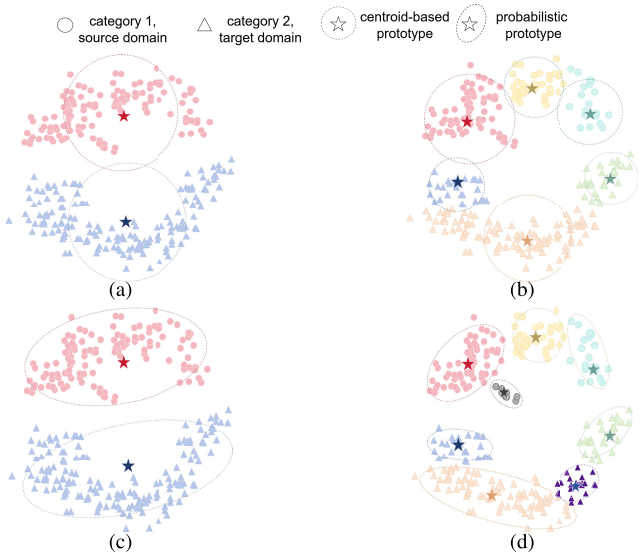


Fig. 2. Comparison of prototypes in the source and target domains. (a) Centroid-based single prototype. (b) Centroid-based multiple prototypes. (c) Probabilistic single prototype. (d) Probabilistic multiple prototypes (ours).

confusion caused by coarse global alignment [16]. In ST-based methods, the feature distance between samples and prototypes provides a robust criterion for obtaining their categories, thereby reducing semantic uncertainty in pseudo-labels and mitigating the influence of noise [17].

Despite these advances, existing prototype-based UDA methods for RSI semantic segmentation still face two major challenges.

1) As shown in Fig. 1(a) and (b), intracategory variation within domains and intercategory similarity across domains lead to complex category distributions, whose accurate representation is crucial for domain alignment. However, existing UDA methods typically represent category distributions using centroid-based prototypes, as shown in Fig. 2(a) and (b), which has two limitations: 1) collapsing continuous distributions into discrete centroids obscures subtle pattern variations within the same category (e.g., buildings with and without striped roof textures), leading to coarse category discrimination; and 2) assuming isotropic dispersion reduces the separability of similar categories [e.g., trees and low vegetation (LV)], resulting in overlapping category boundaries.
2) As shown in Fig. 1(c), the frequencies of the same category vary considerably across domains due to scenario differences. For example, man-made categories tend to appear more frequently in urban scenes, whereas natural categories may dominate in rural scenes, particularly impervious surfaces (ISs) and trees. This mismatch introduces category-prior bias between the source and target domains. However, existing UDA methods typically estimate prototypes under the empirical assumption that category priors are consistent across domains. As a result, for the same category, the prototype learned from the source domain may deviate from its counterpart in the target domain.

To address these challenges, we propose a probabilistic multiple prototypes contrastive alignment (PMPCA) method for UDA in RSI semantic segmentation. Specifically given as follows.

1) To tackle the first challenge, we propose a probabilistic multiple prototype learning (PMPL) module, in which multiple prototypes and probabilistic parameters are jointly used to finely represent category distributions in the source and target domains. In this way, the inherent semantics of each category can be better preserved across domains, as shown in Fig. 2(d).
2) To tackle the second challenge, we propose a prior-weighted contrastive (PWC) constraint by establishing prototype-level contrastive learning with category-frequency weighting between the source and target domains, promoting cross-domain structural alignment while alleviating bias induced by category-prior mismatch.
3) In addition, we design an uncertainty-guided curriculum adaptation (UCA) strategy to suppress noisy pseudo-labels by performing curriculum-style UDA according to scenario difficulty, thereby improving the reliability of PMPL and PWC.

In summary, the main contributions of this work are as follows.

1) A PMPL mechanism is proposed to finely represent category distributions, thereby preserving the semantic integrity of each category across the source and target domains.
2) A category-frequency-weighted contrastive constraint is proposed to adaptively calibrate domain alignment, thereby reducing negative transfer between the source and target domains caused by category-prior bias.
3) A curriculum-style UDA strategy is designed to progressively transfer knowledge from the source domain to the

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target domain, which improves the reliability of pseudo-labels.

4) Extensive experiments on four public RSI datasets with six UDA settings demonstrate that the proposed method achieves state-of-the-art segmentation performance, especially in cross-spectral-band and cross-spatial-region generalization.

II. Related Work

A. UDA for RSI Semantic Segmentation

Existing UDA methods for RSI semantic segmentation can be broadly classified into AT-based and ST-based methods [2].

AT-based methods construct a discriminator that engages in an adversarial game with the segmentation model. This interaction aims to confuse the discriminator, making it unable to distinguish whether the input comes from the source or target domain, thereby reducing domain discrepancies and achieving domain alignment [6]. The adversarial game can be applied at multiple levels, including the input, intermediate, and output levels. At the input level, domain shift is primarily characterized by appearance discrepancies between the source and target domains, such as color tone, brightness, and contrast [7], [18]. These discrepancies result in visual deviations across domains. Therefore, input-level AT-based methods typically employ image translation techniques to align RSI styles across domains [19]. A critical challenge for these methods is preserving the semantic structure of RSIs during image translation [20]. At the intermediate level, domain shift manifests as feature discrepancies, which disrupt representation invariance across domains. Therefore, intermediate-level AT-based methods often employ feature-matching techniques to align representations across domains. Common feature-matching strategies include normalization and distance constraints. Normalization adapts source domain features by aligning their mean and variance with those of the target domain, while distance constraints explicitly quantify feature discrepancies between the source and target domains using metrics, such as Kullback–Leibler (KL) divergence and maximum mean discrepancy (MMD). At the output level, domain shift manifests as misalignment in structural predictions, where the predicted results for the source and target domains differ in category layout and object composition. Therefore, output-level AT-based methods enforce prediction consistency across domains by applying a consistency loss to the output probabilities [8]. However, AT-based methods are often prone to training collapse.

ST-based methods establish a closed-loop process between the teacher and student models, facilitating progressive knowledge transfer from the source to the target domain [10]. The teacher model generates pseudo-labels for target domain RSIs, while the student model is jointly supervised by true labels from source domain RSIs and pseudo-labels from target domain RSIs. In this process, the teacher model parameters are updated using the EMA of the student model parameters, whereas the student model parameters are updated through backpropagation [17].

However, pseudo-labels are inherently susceptible to noise, which may lead to error accumulation in the student model [21]. To mitigate this issue, various pseudo-label enhancement strategies have been proposed, including fixed and dynamic confidence-threshold filtering [11]. These methods retain only the most reliable pseudo-labels derived from pixels with high prediction confidence, thereby suppressing noise. For example, Zhang et al. [22] suggested that the topological relationships among neighboring pixels remain relatively invariant across domains, thereby enhancing the local structural consistency of pseudo-labels. Ran et al. [12] integrated the spatial context of pixels with their prediction confidences to perform weighted refinement of pseudo-labels, thereby reducing noise in low-confidence regions.

B. Prototype Learning for UDA

A prototype is generated by computing the mean of samples belonging to the same category in the feature space, thereby abstracting the common pattern of that category [23]. Accordingly, clustering can be performed based on the feature distances between samples and prototypes, enabling category-level semantics to be represented in an unsupervised manner. Depending on the number of clusters, prototypes can be categorized as either single or multiple [2].

Single-prototype-based methods assume that each category can be adequately represented by a single prototype. For example, Zhu et al. [24] utilized prototypes to store historical features from both the source and target domains and updated them with current features to form domain-invariant representations. While single-prototype-based methods perform well in simple scenarios, they struggle to capture high intracategory variability in more complex scenarios, thereby limiting model generalization. In contrast, multiple-prototype-based methods assume that each category is better represented by multiple prototypes. For example, Fan et al. [4] and Gao et al. [2] proposed representing each category with multiple prototypes to achieve fine-grained domain alignment. Gao et al. [2] further applied multiscale random masking to RSIs, forcing the model to preserve semantics while reconstructing masked regions, thereby enhancing spatial context. Fan et al. [4] suggested that the number of prototypes affects the model's generalization capability and proposed dynamically adjusting the number of prototypes based on the variance within each category.

Prototypes can also be classified as centroid-based or probabilistic. Centroid-based prototypes are computed under Euclidean geometry, such as nearest-neighbor-based and K-means-based formulations, and represent each category by a single point. This formulation implicitly assumes an isotropic category distribution, i.e., equal variance in all directions. In contrast, probabilistic prototypes are computed based on distribution densities, e.g., Gaussian distributions, allowing each prototype to be anisotropic and to have its own shape. In contrast, probabilistic prototypes are more flexible. Therefore, in the field of natural images, centroid-based prototypes have been extended to probabilistic prototypes to address intracategory variability and intercategory similarity. For example, Xie et al. [10] suggested that pixels from the same category share a similar distribution despite exhibiting variability and are domain-independent. In contrast, Hao et al. [21] argued that such similarity may lead to blurred decision boundaries

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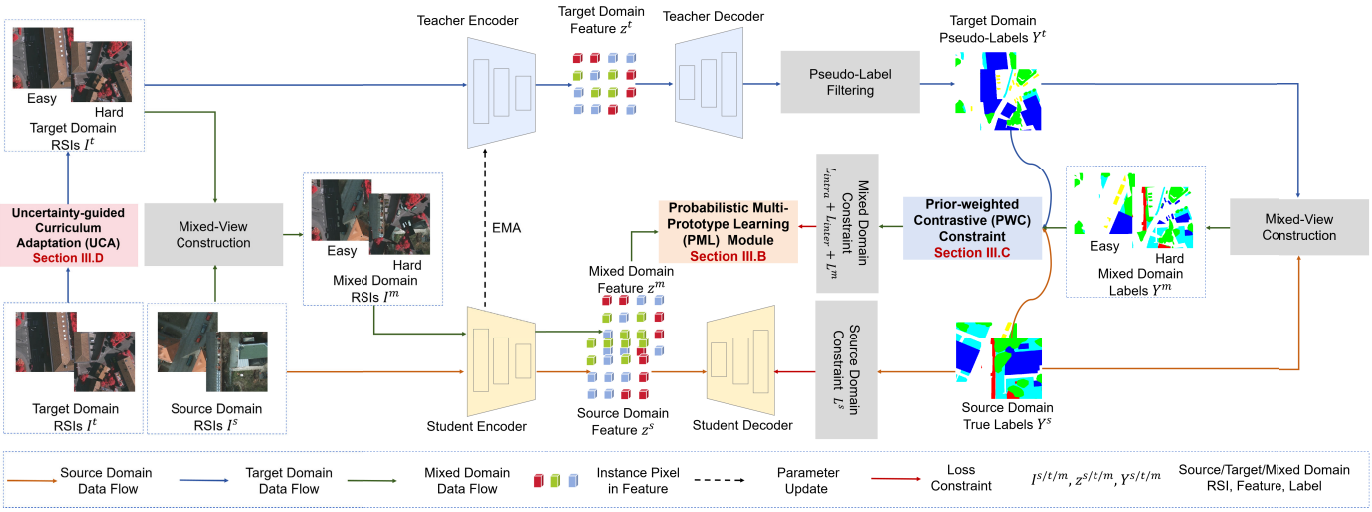


Fig. 3. Overview of the proposed method.

and thus proposed a probabilistic single prototype for each category. Meanwhile, Yu et al. [25] utilized both centroid-based and probabilistic multiple prototypes to jointly represent each category.

In the field of RSI, this work is the first to propose probabilistic multiple prototypes for representing category distributions. Unlike existing methods for natural images, this work considers both category-conditional alignment and category-prior alignment, thereby enhancing cross-domain alignment in RSI semantic segmentation.

III. Methodology

A. Task Definition and Method Overview

Formally, the source domain is denoted as $\mathcal{D}^s = \{(I_i^s, Y_i^s)\}_{i=1}^{N^s}$, where $I_i^s \in \mathbb{R}^{H\times W\times 3}$ denotes the ith labeled source domain RSI, and $Y_i^s \in \{1,\ldots,C\}^{H\times W}$ represents its corresponding category label in one-hot coding format. The target domain is denoted as $\mathcal{D}^t = \{I_j^t\}_{j=1}^{N^t}$, where $I_j^t \in \mathbb{R}^{H\times W\times 3}$ are unlabeled target domain RSI. Here, H and W are the height and width of the RSIs, respectively; N^s and N^t are the number of RSIs in $\mathcal{D}^s$ and $\mathcal{D}^t$, respectively; C is the total number of categories; and i and j are image indexes. The objective of UDA is to effectively transfer knowledge learned from $\mathcal{D}^s$ to $\mathcal{D}^t$ through domain alignment, thereby enhancing the model's generalization ability and improving its performance on $\mathcal{D}^t$.

As illustrated in Fig. 3, PMPCA consists of four components: a teacher–student architecture, the PMPL module, the PWC constraint, and the UCA strategy. The teacher model generates pseudo-labels $Y^t \in \{1,\ldots,C\}^{H\times W}$ for target domain RSIs $I^t \in \mathcal{D}^t$, while the student model is trained using source domain RSIs $I^s \in \mathcal{D}^s$ with true labels Y^s and mixed labels $Y^m \in \{1,\ldots,C\}^{H\times W}$ from the mixed domain. The teacher model parameters are updated as the EMA of the student model parameters, whereas the student model parameters are updated through gradient backpropagation. Built upon the teacher–student architecture, the PMPL module learns shared probabilistic multiple prototypes from mixed domain views; the PWC constraint regularizes cross-domain alignment in the prototype space, and the UCA strategy schedules target domain scenarios from easy to hard during training. The details of each component are introduced in Sections III-B–III-D.

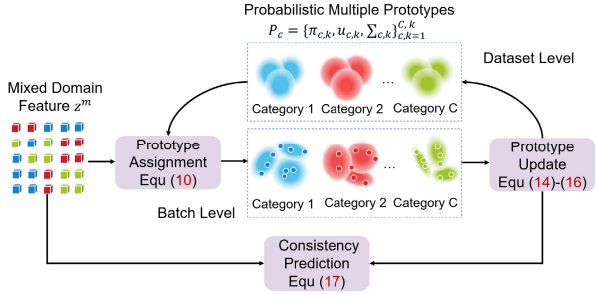


Fig. 4. Schematic of PMPL module.

B. PMPL Module

We propose the PMPL module. For each category, multiple prototypes are constructed to capture diverse instance patterns (e.g., variations in scale, texture, and color) within the same category, thereby improving tolerance to intracategory variability. For each pattern, probabilistic parameters (i.e., the mean and covariance) are further introduced to jointly capture central tendency and dispersion shape, which helps reduce boundary overlap between similar categories. In this way, the distribution of each category can be represented in a fine-grained manner in both the source and target domains, thereby better preserving category semantics across domains.

The PMPL module consists of mixed-view construction, prototype assignment, prototype update, and consistency prediction, as illustrated in Fig. 4.

1) Mixed-View Construction: By forcing the model to adapt to coherent regions originating from different domains within a single RSI, the category-mix operation [36] encourages the construction of domain-shared prototypes. Specifically, a subset of categories $C_s \subset C$ is randomly selected. A binary mask $M \in \{0,1\}^{H\times W}$ is then generated, where instance pixels belonging to the selected categories $c \in C_s$ are set to 1, while the remaining instance pixels are set to 0. The mask M is

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applied to blend the source domain RSI I^s and target domain RSI I^t, yielding a mixed domain RSI I^m

$$I^m = M \odot I^s + (1 - M) \odot I^t \tag{1}$$

where $\odot$ denotes the elementwise multiplication, and $M(x) = 1$ for instance pixels in the selected categories C_s and $M(x) = 0$ otherwise. Accordingly, the label Y^m of I^m is generated by combining the source domain ground-truth label Y^s and the target domain pseudo-label Y^t.

2) Prototype Assignment: Inspired by the universal approximation property of Gaussian mixture distributions [23], [26], [27], [28], the distribution of each category is modeled as a weighted combination of multiple Gaussian components, where each component corresponds to a prototype of that category with probabilistic parameters

$$\mathcal{P}_{c,k}(z) = \mathcal{N}\left(z; \mu_{c,k}, \Sigma_{c,k}\right) \tag{2}$$

where $c = 1, 2, \ldots, C$ and $k = 1, 2, \ldots, K$. z is an instance pixel of the mixed-domain feature. K is the number of prototypes for category c. The Gaussian distribution is defined as

$$\mathcal{N}(z; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(z - \mu)^\top \Sigma^{-1} (z - \mu)\right)$$

with mean vector $\mu_{c,k} \in \mathbb{R}^D$ and covariance matrix $\Sigma_{c,k} \in \mathbb{R}^{D \times D}$, D is feature dimension.

The probabilistic multiple prototypes for category c are represented as $\mathcal{P}_c = \{P_{c,1}, P_{c,2}, \ldots, P_{c,K}\}$, and their relationship to a single prototype is formulated as

$$\mathcal{P}_c(z) = \sum_{k=1}^{K} \pi_{c,k} \mathcal{N}\left(z; \mu_{c,k}, \Sigma_{c,k}\right) \tag{3}$$

where $\pi_{c,k}$ is the weight of the kth prototype and satisfies $\sum_{k=1}^{K} \pi_{c,k} = 1$.

The assignment of each instance pixel to the prototypes of its category is performed via an online expectation step. The assignment weight $\gamma_{c,k}(z)$ denotes the probability that the feature z of an instance pixel belongs to the kth prototype of category c

$$\gamma_{c,k}(z) = \frac{\pi_{c,k} \mathcal{N}\left(z; \mu_{c,k}, \Sigma_{c,k}\right)}{\sum_{j=1}^{K} \pi_{c,j} \mathcal{N}\left(z; \mu_{c,j}, \Sigma_{c,j}\right)}. \tag{4}$$

This ensures that $\{\gamma_{c,k}(z)\}_{k=1}^{K}$ forms a valid probability distribution for each instance pixel, i.e., $\sum_{k=1}^{K} \gamma_{c,k}(z) = 1$.

However, direct soft assignment may lead to prototype degradation, where some prototypes are over-utilized by dense instance pixels, while others are under-utilized by sparse instance pixels. To alleviate this imbalance, we introduce an entropy-regularized optimal transport (OT) constraint [29] to enforce a more balanced assignment between instance pixels and prototypes. The assignment cost $A_{c,k}$ between an instance pixel feature z and the kth prototype of category c is defined as the negative log-likelihood

$$A_{c,k} = -\log \mathcal{N}\left(z; \mu_{c,k}, \Sigma_{c,k}\right). \tag{5}$$

This cost is minimized when the instance pixel feature closely matches the prototype, thereby encouraging high-probability assignments. The entropy-regularized OT problem is then formulated as

$$\min_{Q \in \mathcal{Q}} \langle Q, A \rangle_F - \varepsilon H(Q) \tag{6}$$

where $Q \in \mathbb{R}_+^{L \times K}$ is the transport matrix, L is the number of instance pixels, and K is the number of prototypes. The term $\langle Q, A \rangle_F = \sum_{l=1}^{L} \sum_{k=1}^{K} Q_{l,k} A_{l,k}$ is the Frobenius inner product between the transport matrix Q and the cost matrix A, representing the total transport cost. The scalar ε controls the strength of entropy regularization, and $H(Q)$ is the entropy of Q, defined as $H(Q) = -\sum_{l=1}^{L} \sum_{k=1}^{K} Q_{l,k} \log(Q_{l,k})$.

The feasible set $\mathcal{Q}$ imposes row and column marginal constraints

$$\mathcal{Q} = \left\{ Q \in \mathbb{R}_+^{L \times K} \middle| Q 1_K = \frac{1}{L} 1_L,\ Q^\top 1_L = \frac{1}{K} 1_K \right\} \tag{7}$$

where 1_L and 1_K are all-ones vectors of sizes L and K, respectively. The row constraint $Q1_K = 1/L 1_L$ balances the total transport mass assigned to each instance pixel, while the column constraint $Q^\top 1_L = 1/K 1_K$ ensures that each prototype receives the same total mass, thereby encouraging uniform prototype utilization. To efficiently obtain the OT plan, Sinkhorn scaling is adopted [2]. Let $G = \exp(-A/\varepsilon)$ be the Gibbs kernel defined from the cost matrix A, where the exponential is applied elementwise. The approximate optimal assignment matrix $Q^\star$ is then given by

$$Q^\star = \operatorname{diag}(u)\, G \operatorname{diag}(v) \tag{8}$$

where $u \in \mathbb{R}^L$ and $v \in \mathbb{R}^K$ are scaling vectors iteratively updated to satisfy the marginal constraints. $\operatorname{diag}(u)$ and $\operatorname{diag}(v)$ are diagonal matrices constructed from u and v, respectively. Sinkhorn scaling is performed by alternating the following updates:

$$u^{(q+1)} \leftarrow \frac{1}{L} 1_L \oslash \left(G v^{(q)}\right), \quad v^{(q+1)} \leftarrow \frac{1}{K} 1_K \oslash \left(G^\top u^{(q+1)}\right) \tag{9}$$

where q is the iteration index and $\oslash$ denotes elementwise division. This iterative procedure yields a balanced assignment $Q^\star$ that both respects the cost structure and avoids prototype collapse.

Finally, the assignment weights $\gamma_{c,k}^{(l)}$ are normalized as

$$\gamma_{c,k}^{(l)} = \frac{Q_{l,k}^\star}{\sum_{j=1}^{K} Q_{l,j}^\star} \tag{10}$$

where $\gamma_{c,k}^{(l)}$ is the normalized assignment weight between the lth instance pixel and the kth prototype of category c. $Q_{l,k}^\star$ is the entry of the OT matrix $Q^\star$, representing the unnormalized transport mass from the lth instance pixel to the kth prototype. $\sum_{j=1}^{K} Q_{l,j}^\star$ normalizes the assignment weights for each instance pixel, ensuring that the assignments across all prototypes sum to 1 for every instance pixel, i.e., $\sum_{k=1}^{K} \gamma_{c,k}^{(l)} = 1$.

3) Prototype Update: Once the assignment weights $\gamma_{c,k}^{(l)}$ are obtained from the expectation step, the batch-level sufficient statistics for each prototype of category c, including the effective instance pixel count $n_{c,k}$, mean vector $\bar{\mu}_{c,k}$, and covariance matrix $\bar{\Sigma}_{c,k}$, are computed in a maximization step

$$n_{c,k} = \sum_{l=1}^{L} \gamma_{c,k}^{(l)} \tag{11}$$

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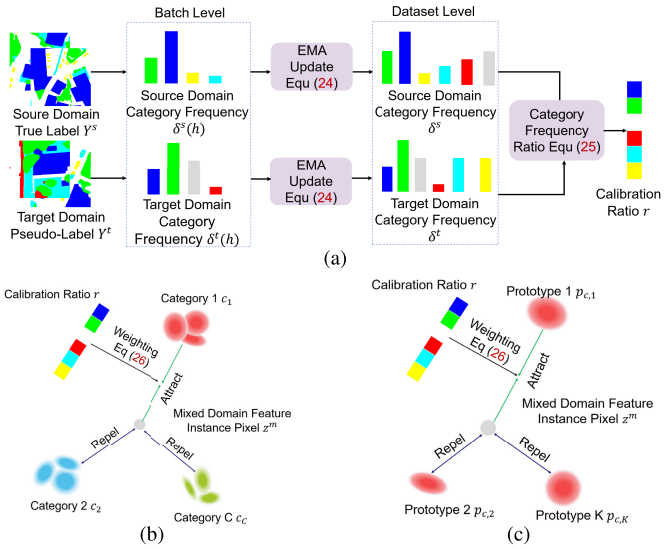


Fig. 5. Schematic of PWC constraint. (a) Domain-specific category frequency estimation. (b) Intracategory contrastive constraint. (c) Intercategory contrastive constraint.

$$\bar{\mu}_{c,k} = \frac{1}{n_{c,k}} \sum_{l=1}^{L} \gamma_{c,k}^{(l)} z \tag{12}$$

$$\bar{\Sigma}_{c,k} = \frac{1}{n_{c,k}} \sum_{l=1}^{L} \gamma_{c,k}^{(l)} \left(z - \bar{\mu}_{c,k}\right)\left(z - \bar{\mu}_{c,k}\right)^{\top}. \tag{13}$$

To preserve temporal stability and avoid noisy updates, we apply an EMA strategy to update the parameters of each prototype at the dataset level, denoted as $\phi_{c,k} = \{\pi_{c,k}, \mu_{c,k}, \Sigma_{c,k}\}$. The updated prototype parameters $\phi_{c,k}$ are computed as

$$\mu_{c,k} \leftarrow \beta \mu_{c,k} + (1-\beta)\bar{\mu}_{c,k} \tag{14}$$

$$\Sigma_{c,k} \leftarrow \beta \Sigma_{c,k} + (1-\beta)\bar{\Sigma}_{c,k} \tag{15}$$

$$\pi_{c,k} \leftarrow \beta \pi_{c,k} + (1-\beta)\frac{n_{c,k}}{L} \tag{16}$$

where $\beta \in [0, 1)$ is the momentum coefficient controlling the update rate, empirically set to 0.999 [10].

4) Consistency Prediction: To generate consistent predictions across the source and target domains, the supervision on the mixed domain RSI I^m is defined as

$$\mathcal{L}^m = -\frac{1}{H \times W} \sum_{j=1}^{H \times W} \sum_{c=1}^{C} w_p \mathbb{1}\left[Y_j^m = c\right] \log P_{j,c}^m \tag{17}$$

where $\mathcal{L}_m$ is a consistency prediction constraint in the mixed domain based on cross-entropy. P^m is the predicted probability map produced by the student model by applying the softmax function to the logits.

C. PWC Constraint

We propose the PWC constraint, as illustrated in Fig. 5. Specifically, instance-prototype contrastive learning is established between the source and target domains. Each instance is attracted to the nearest prototype of its corresponding category while being repelled from prototypes of other categories. In this way, intracategory compactness and intercategory separability are enhanced, thereby promoting structural alignment across domains. Furthermore, the category frequencies of the source and target domains are estimated as prior weights, which helps mitigate bias induced by category-prior mismatch and further refine the structural alignment process.

The PWC constraint consists of an intracategory contrastive constraint, an intercategory contrastive constraint, and category-prior estimation and weighting.

1) Intracategory Contrastive Constraint: For instance, for pixel z of a mixed-domain feature belonging to category c, its nearest (positive) prototype $\mathcal{P}^+ = p_{c,k_c^*}$ is selected from K prototypes of that category as the one with the lowest negative log-likelihood

$$k_c^* = \arg\min_{k \in \{1,2,\ldots,K\}} \left(-\log \mathcal{N}\left(z; \mu_{c,k}, \Sigma_{c,k}\right)\right) \tag{18}$$

where k_c^* is the index of the nearest prototype. $p_{c,k}$ is the kth prototype of category c. Both the instance pixel feature z and the prototype $p_{c,k}$ are ℓ_2-normalized. Accordingly, The remaining prototypes of category c are treated as negatives and denoted by $\mathcal{P}_c^- = \{p_{c,k} \mid k \in \{1, \ldots, K\}, k \neq k_c^*\}$.

The intracategory contrastive constraint $\mathcal{L}_{\text{intra}}$ is then defined as instance pixel-to-prototype contrastive learning within the same category, as shown in Fig. 5(b)

$$\mathcal{L}_{\text{intra}} = -\log \frac{\exp\left(\left\langle z, \mathcal{P}^+ \right\rangle / \tau\right)}{\exp\left(\left\langle z, \mathcal{P}^+ \right\rangle / \tau\right) + \sum_{\mathcal{P}^- \in \mathcal{P}_c^-} \exp\left(\left\langle z, \mathcal{P}^- \right\rangle / \tau\right)} \tag{19}$$

where $\tau > 0$ is the temperature parameter that controls the sharpness of the contrastive distribution.

This encourages the instance pixel z to be closer to its nearest prototype $\mathcal{P}^+$ and farther from the other prototypes $\mathcal{P}^+$ of the same category. As a result, instance pixels sharing similar patterns are clustered around their corresponding prototypes, which enhances intracategory compactness.

2) Intercategory Contrastive Constraint: The distance s_c from instance pixel feature z in the mixed domain RSI to category c is defined via the nearest prototype k_c^* of that category

$$s_c = -\log \mathcal{N}\left(z; \mu_{c,k_c^*}, \Sigma_{c,k_c^*}\right) \tag{20}$$

where the instance pixel feature z and the prototype $p_{c,k}$ are ℓ_2-normalized. Based on the set of distances $\{s_c\}_{c=1}^C$, the predicted category of z is

$$c^* = \arg\min_{c \in \{1,2,\ldots,C\}} s_c \tag{21}$$

where c^* is the predicted category index.

The intercategory contrastive constraint $\mathcal{L}_{\text{inter}}$ is formulated as instance pixel-to-prototype contrastive learning across different categories, as shown in Fig. 5(c)

$$\mathcal{L}_{\text{inter}} = -\log \frac{\exp\left(-s_{c^*}\right)}{\sum_{c=1}^{C} \exp\left(-s_c\right)} \tag{22}$$

where c indexes the candidate categories.

This decreases the distance s_{c^*}, pulling instance pixel z closer to the prototype of its predicted category c^*, while increasing the distances s_c for all $c \neq c^*$, thereby pushing z away from prototypes of other categories and strengthening intercategory separability.

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3) Category-Prior Estimation and Weighting: The category frequency of category c in the source domain $\mathcal{D}^s$ and target domain $\mathcal{D}^t$ at iteration h is estimated as follows:

$$\delta_c^s(h) = \frac{1}{V^s}\sum_{i\in I^s}\mathbb{1}\left(Y_i^s = c\right), \quad \delta_c^t(h) = \frac{1}{V^t}\sum_{i\in I^t}\mathbb{1}\left(Y_i^t = c\right) \tag{23}$$

where $\delta_c^s(h)$ and $\delta_c^t(h)$ are the batch-level category frequencies of the source and target domains, respectively. $\mathbb{1}(\cdot)$ is the indicator function. V^s and V^t are the numbers of instance pixels from the source and target domains in the batch, respectively. Y_i^s and Y_i^t are the corresponding labels.

The dataset-level category frequencies are then updated using an EMA to incorporate both previous and current estimates, as shown in Fig. 5(a)

$$\delta_c^s \leftarrow \eta\delta_c^s + (1-\eta)\,\delta_c^s(h), \quad \delta_c^t \leftarrow \eta\delta_c^t + (1-\eta)\,\delta_c^t(h) \tag{24}$$

where η is the momentum coefficient controlling the update rate, empirically set to 0.999 [10].

To address the mismatch in category frequencies between the source and target domains, the empirical calibration ratio r_c is defined as

$$r_c = \frac{\delta_c^s + \varepsilon}{\delta_c^t + \varepsilon} \tag{25}$$

where $\varepsilon > 0$ is a small constant for numerical stability. The calibration weight for each instance pixel is then defined as

$$W_i = M_i \cdot 1 + (1 - M_i)\cdot r_c \tag{26}$$

where $M_i = 1$ indicates that the ith instance pixel comes from the source domain and contributes to the loss with a unified weight, while $M_i = 0$ indicates that it comes from the target domain and is weighted by the calibration ratio r_c corresponding to its category.

This calibration ratio adjusts the contribution of different categories in the target domain based on their relative frequencies in the source domain. Specifically, if a category is more frequent in the target domain than in the source domain, its calibration ratio r_c becomes smaller, and the weight assigned to the corresponding target instance pixels is reduced. Finally, the intracategory and intercategory contrastive losses are weighted by the calibration weight as follows:

$$\mathcal{L}_{\text{intra}} \leftarrow W\mathcal{L}_{\text{intra}}, \quad \mathcal{L}_{\text{inter}} \leftarrow W\mathcal{L}_{\text{inter}} \tag{27}$$

where W is the calibration weight for each instance pixel. The calibration weight W is dynamically computed from the estimated source domain and target domain category-frequency ratio.

D. UCA Strategy

The complex category distributions and category-prior mismatch may lead to incorrect predictions in the target domain, thereby increasing the risk of generating noisy pseudo-labels. Inspired by the human cognitive principle of learning from easy to hard [6], we propose the UCA strategy, as illustrated in Fig. 6. This strategy evaluates the difficulty of target domain scenarios based on prediction uncertainty, enabling the model to first learn from easy scenarios and then gradually incorporate hard scenarios. In this way, the progressive UDA process helps reduce the influence of noisy pseudo-labels and facilitates more accurate knowledge transfer across domains.

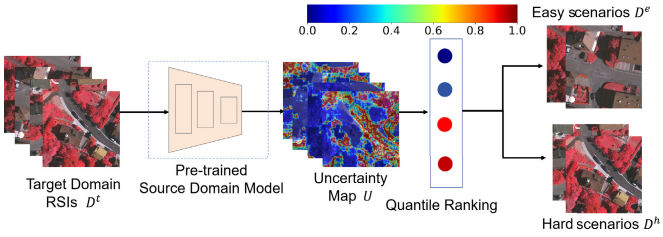


Fig. 6. Schematic of UCA strategy.

1) Uncertainty Estimation: A well-trained model in the source domain can provide effective uncertainty estimates for the target domain. The uncertainty is defined by information entropy, which is formulated as

$$u = H\left(P\left(y \mid z\right)\right) = -\sum_{c=1}^{C} P\left(y = c \mid z\right)\ \log P\left(y = c \mid z\right) \tag{28}$$

where z is an instance pixel from the target domain feature, and H is the entropy function.

Correspondingly, the uncertainty of a target domain RSI $U(I^t)$ is obtained by averaging the uncertainty over all instance pixels

$$U\left(I^t\right) = \frac{1}{N}\sum_{i=1}^{H}\sum_{j=1}^{W} u_{i,j} \tag{29}$$

where $N = H \times W$ is the number of instance pixels. $U(I^t)$ reflects the prediction uncertainty of the target domain RSI. To obtain comparable uncertainty values across RSIs, we normalize $U(I^t)$ as

$$\bar{U}\left(I^t\right) = \frac{U\left(I^t\right) - U_{\min}}{U_{\max} - U_{\min}} \tag{30}$$

where $U_{\min}$ and $U_{\max}$ denote the minimum and maximum values of $U(I^t)$ over the target domain, respectively. A higher value of $\bar{U}(I^t)$ indicates a harder scenario, in which the model is less confident in its predictions. Conversely, a lower value indicates an easier scenario, in which the model is more confident in its predictions.

To guide the model's learning, the target domain RSIs are ranked in ascending order according to their uncertainty

$$\mathcal{D}^e = \left\{I^t \in \mathcal{D}^t \mid \bar{U}\left(I^t\right) \le \text{Quantile}_\gamma\left(\left\{\bar{U}\left(I^t\right) \mid I^t \in \mathcal{D}^t\right\}\right)\right\} \tag{31}$$

where $\mathcal{D}^e(\gamma)$ denotes the set of easy-to-adapt RSIs, $\gamma \in (0, 1]$ is the quantile, and $\text{Quantile}_\gamma(\cdot)$ denotes the γ-quantile operator over the uncertainty values of all target domain RSIs.

2) Curriculum Design: The training process is divided into two phases.

1) In the warm-up phase, only easy-to-adapt RSIs $I^t \in \mathcal{D}^e$ with low uncertainty are selected for training. Pseudo-labels Y^t are generated only for these selected RSIs, which are then combined with source domain RSIs.
2) In the enhancement phase, hard-to-adapt RSIs $I^t \in \mathcal{D}^h$, where $\mathcal{D}^h \cup \mathcal{D}^e = \mathcal{D}^t$, are introduced for training. The model progressively adapts to these more challenging scenarios, thereby enhancing its generalization capability.

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E. Optimization Objective of PMPCA

The constraint on the source domain is defined as

$$\mathcal{L}^s = -\frac{1}{H \times W} \sum_{i=1}^{H \times W} \sum_{c=1}^{C} \mathbb{1}\left[Y_i^s = c\right] \log P_{i,c}^s \tag{32}$$

where P^s is the predicted probability map obtained by applying the softmax function to the logits of the student model. Therefore, the overall optimization objective of PMPCA is defined as

$$\mathcal{L} = \mathcal{L}^s + \lambda\left(\mathcal{L}^m + \mathcal{L}_{\text{intra}} + \mathcal{L}_{\text{inter}}\right) \tag{33}$$

where $\mathcal{L}^m$ denotes the consistency constraint on the mixed domain, and $\mathcal{L}_{\text{intra}}$ and $\mathcal{L}_{\text{inter}}$ denote the prior-weighted intra-category and intercategory contrastive constraints, respectively. λ is the balancing factor. Algorithm 1 presents the training pipeline of PMPCA, which adopts a two-stage curriculum-based data scheduling mechanism. Both stages are optimized under the same optimization objective in (33).

Algorithm 1 Training Pipeline of PMPCA

Input: source domain $\mathcal{D}^s$, target domain $\mathcal{D}^t$, student model f_θ, teacher model f_ξ, probabilistic multiple prototypes $\mathcal{P}$
Output: optimized student model parameters θ^*
for each $I^t \in \mathcal{D}^t$ **do**
  Obtain target prediction $\bar{P}^t$ using source domain-pretrained f_θ with Eq. (28);
  Compute uncertainty map $U(I^t)$ with Eq. (29);
Split $\mathcal{D}^t$ into easy subset $\mathcal{D}^e$ and hard subset $\mathcal{D}^h$ with Eqs. (30)–(31);
for $e = 1, 2, \ldots, E$ **do**
  Sample a mini-batch B^s from $\mathcal{D}^s$;
  if $e \leq E/2$ **then**
    Sample an easy mini-batch B^t from $\mathcal{D}^e$;
  else
    Sample a hard mini-batch B^t from $\mathcal{D}^h$;
  Obtain teacher prediction P^t and pseudo-label Y^t;
  Construct mixed domain view I^m and mixed label Y^m with Eq. (1);
  Obtain mixed domain feature Z^m and student prediction P^m;
  Obtain student prediction P^s;
  for each category c **do**
    Extract category-specific mixed domain features Z_c^m from Z^m;
    Compute probabilistic prototype assignment weights $\gamma_{c,k}^{(l)}$ with Eqs. (5)–(10);
    Update category-specific probabilistic multiple prototypes $\mathcal{P}_c$ with Eqs. (11)–(16);
  Compute the supervised constraint $\mathcal{L}^s$ on the source domain and the consistency constraint $\mathcal{L}^m$ on the mixed domain with Eq. (32) and (17);
  Estimate batch-level and dataset-level category frequencies δ_c^s and δ_c^t with Eqs. (23)–(24);
  Compute the calibration ratio r_c and instance-level calibration weight W_i with Eqs. (25)–(26);
  Compute the prior-weighted intra-category contrastive loss $\mathcal{L}_{\text{intra}}$ and inter-category contrastive loss $\mathcal{L}_{\text{inter}}$ with Eqs. (18)–(19), (20)–(22), (27);
  Compute the total loss $\mathcal{L}$ with Eq. (33);
  Update θ with $\mathcal{L}$;
  Update ξ with EMA;
return θ^*;

IV. Experiments

A. Dataset and Implementation Details

1) Dataset: Experiments are conducted on four publicly available RSI datasets: ISPRS-Vaihingen (VAI), ISPRS-Potsdam (POT) [30], LoveDA-Urban (URB), and LoveDA-Rural (RUR) [31]. The POT and VAI datasets were collected from the geographic regions of Potsdam and Vaihingen in Germany using an airborne platform and contain six RSI object categories: clutter, IS, building, LV, tree, and car. The POT dataset includes infrared–red–blue (IRRG) and red–green–blue (RGB) modes, whereas VAI includes only the IRRG mode. All RSIs are cropped into patches of 512×512 pixels with a stride of 512 pixels, resulting in 1696 and 4598 RSIs for the POT and VAI datasets, respectively. The URB and RUR datasets were collected from the geographic regions of Nanjing, Changzhou, and Wuhan in China using a satellite platform and contain seven RSI object categories: background, building, road, water, barren, forest, and agriculture. Both datasets use the RGB mode. Similarly, all RSIs are cropped into patches of 512×512 pixels with a stride of 512 pixels, resulting in 4624 and 2708 RSIs for the URB and RUR datasets, respectively.

Six UDA settings are designed to evaluate the model's generalizability from the source domain to the target domain: POT-IRRG → VAI, POT-RGB → VAI, VAI → POT-IRRG, VAI → POT-RGB, RUR → URB, and URB → RUR.

2) Implementation Details: Experiments are conducted on an Ubuntu 22.04 LTS operating system with an NVIDIA GeForce RTX 4090 GPU and CUDA 11.0, and the proposed method is implemented using the MMSegmentation toolbox with MMCV 1.3.7, PyTorch 1.7, and Python 3.8. The batch size is set to 4, and the initial learning rate is 6×10^{-5}. The learning rate schedule consists of 1.5k warm-up iterations, followed by polynomial decay over a total of 40k iterations. The AdamW optimizer is adopted with momentum parameters of 0.9 and 0.999, and a weight decay of 0.01. All RSIs are normalized using the mean [123.675, 116.28, 103.53] and standard deviation [58.395, 57.12, 57.375].

Weak data augmentation strategies include random cropping with a maximum ratio of 0.75, random resizing with a scale ratio ranging from 0.5 to 2.0, and random flipping with a probability of 0.5. Strong data augmentation strategies include category mixing and color jitter with an intensity and probability of 0.2. Both the teacher and student models use Mix Transformer-B5 as the encoder with ImageNet-pretrained weights and DAFormer [36] as the decoder with an ASPP fusion module. The pseudo-label threshold α is set to 0.968. The prototype update momentum β is set to 0.999, and each category is represented by seven prototypes with 128 dimensions ($K = 7, D = 128$). The temperature parameter τ and balancing factor λ are set to 0.1 and 1.0, respectively. The quantile γ is set to 0.25, corresponding to the Q1 quartile.

B. Evaluation Metrics and Comparison Methods

1) Evaluation Metrics: The standard performance evaluation metrics for semantic segmentation include intersection over union (IoU) and accuracy (ACC) [39], [40].

IoU measures the overlap between the predicted and labeled regions for each category, whereas ACC measures the

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TABLE I

CROSS-DOMAIN SEMANTIC SEGMENTATION PERFORMANCE OF DIFFERENT UDA METHODS ON THE ISPRS DATASET UNDER THE POT-IRRG → VAI SETTING. RESULTS ARE REPORTED AS IOU/ACC. BEST VALUE IN BOLD AND SECOND BEST VALUE IN UNDERLINE

Method	clutter	impervious surface	car	tree	low vegetation	building	avg
MemoryNet (TGRS'23)	21.07/72.44	63.45/76.42	40.20/53.47	53.28/62.05	43.85/62.85	70.48/88.86	48.72/69.35
UDARS (RS'23)	38.95/46.74	75.62/85.41	47.90/93.81	65.56/69.83	60.74/81.82	87.20/95.63	62.66/78.87
PiPa (ACM MM'23)	39.26/62.21	77.44/89.27	59.52/81.34	71.98/85.26	57.90/66.93	87.47/95.92	65.59/80.15
SePico (TPAMI'23)	59.71/**92.69**	76.18/82.81	50.84/94.73	69.96/93.63	50.73/54.88	85.21/96.30	65.44/85.84
ST-DASeg (JAG'24)	38.13/80.34	74.41/78.53	51.60/86.91	72.79/82.20	**62.90**/79.80	85.74/**97.10**	64.26/84.15
PPPC (NN'25)	14.63/53.62	76.75/86.38	60.20/83.05	59.12/62.66	56.15/**82.77**	87.28/92.92	59.02/76.90
ERF (TASE'25)	45.55/59.61	77.54/85.01	53.77/82.17	71.42/88.59	56.34/65.19	84.91/95.64	64.92/79.37
UDASS (TPAMI'25)	36.67/58.69	79.15/86.82	61.98/91.97	73.10/91.48	57.71/63.61	87.58/97.07	66.03/81.61
MEBS (TGRS'25)	64.75/91.02	76.49/83.54	39.21/**96.46**	70.23/93.22	51.09/55.30	**87.71**/96.22	64.91/85.96
MCDA (TGRS'25)	45.90/58.61	78.11/**89.64**	56.82/78.21	71.63/90.15	55.00/60.84	86.48/94.81	65.66/78.71
DistilledDA [37] (WACV'26)	60.68/91.03	77.30/85.28	49.92/66.44	67.60/**94.30**	48.06/53.75	84.92/92.60	64.75/80.57
UCDS [38] (TIP'26)	42.43/61.06	**79.64**/87.52	**63.04**/79.49	73.47/90.51	58.81/65.89	86.82/96.84	67.37/80.22
PMPCA (ours)	**68.58**/90.90	76.75/83.68	48.50/95.19	**73.64**/91.76	58.43/64.82	87.10/96.56	**68.83/87.15**

TABLE II

CROSS-DOMAIN SEMANTIC SEGMENTATION PERFORMANCE OF DIFFERENT UDA METHODS ON THE ISPRS DATASET UNDER THE VAI → POT-IRRG SETTING. RESULTS ARE REPORTED AS IOU/ACC. BEST VALUE IN BOLD AND SECOND BEST VALUE IN UNDERLINE

Method	clutter	impervious surface	car	tree	low vegetation	building	avg
MemoryNet (TGRS'23)	19.88/22.88	61.41/74.72	48.09/52.53	38.37/59.02	52.74/68.22	61.71/83.97	47.03/60.22
UDARS (RS'23)	11.81/12.74	71.66/85.35	62.76/90.54	58.55/71.48	68.62/82.65	76.85/95.72	58.37/73.08
PiPa (ACM MM'23)	**39.72/61.38**	74.42/83.55	70.37/89.02	60.00/69.78	69.59/89.33	80.60/95.77	60.28/72.36
SePico (TPAMI'23)	27.59/30.90	62.99/86.42	73.12/86.20	53.10/57.90	70.74/84.52	78.30/95.25	60.97/73.53
ST-DASeg (JAG'24)	4.40/4.49	65.41/74.64	**74.56**/91.25	51.15/55.45	67.01/**92.52**	74.66/94.29	56.20/68.77
PPPC (NN'25)	3.07/3.16	63.78/76.58	70.57/87.64	34.96/47.39	61.23/81.84	70.10/91.69	50.62/64.72
ERF (TASE'25)	6.41/6.46	72.64/80.99	70.47/92.55	57.76/72.28	70.79/87.59	78.84/**96.80**	59.49/72.78
UDASS (TPAMI'25)	3.91/3.92	72.49/88.39	64.35/92.38	54.11/59.54	70.04/89.45	80.59/95.82	57.58/71.58
MEBS (TGRS'25)	9.75/10.51	68.20/86.82	69.58/**95.56**	58.28/70.59	69.07/83.85	80.16/92.46	59.17/71.25
MCDA (TGRS'25)	7.68/7.81	74.45/84.98	68.58/89.51	**64.64/89.46**	67.08/78.14	83.50/96.48	60.99/**74.40**
DistilledDA [37] (WACV'26)	4.43/4.46	74.36/87.44	60.88/89.73	61.16/73.00	68.85/86.29	80.53/92.65	58.37/72.26
UCDS [38] (TIP'26)	2.10/2.11	74.41/83.03	70.56/92.85	62.90/72.76	70.23/89.99	81.76/96.13	60.33/72.81
PMPCA (ours)	10.73/10.85	**74.48/91.55**	73.51/90.71	54.70/61.03	**71.05**/89.75	**86.03**/95.68	**61.75**/73.26

TABLE III

CROSS-DOMAIN SEMANTIC SEGMENTATION PERFORMANCE OF DIFFERENT UDA METHODS ON THE ISPRS DATASET UNDER THE POT-RGB → VAI SETTING. RESULTS ARE REPORTED AS IOU/ACC. BEST VALUE IN BOLD AND SECOND BEST VALUE IN UNDERLINE

Method	clutter	impervious surface	car	tree	low vegetation	building	avg
MemoryNet (TGRS'23)	7.55/63.47	57.30/75.69	42.69/58.24	43.38/53.71	32.89/49.86	65.18/78.76	41.50/63.29
UDARS (RS'23)	30.29/54.93	73.22/77.99	49.67/76.00	55.12/70.00	41.21/56.87	77.25/96.19	54.46/72.00
PiPa (ACM MM'23)	47.09/73.25	72.35/**83.23**	60.66/82.30	70.91/87.10	49.99/61.96	85.48/93.59	64.41/80.24
SePico (TPAMI'23)	57.86/92.15	62.75/83.17	50.31/92.87	68.44/89.08	26.51/29.59	84.43/96.77	58.39/80.61
ST-DASeg (JAG'24)	21.98/60.72	69.84/78.63	51.43/86.13	71.73/81.69	53.73/**70.78**	86.08/95.06	59.13/78.83
PPPC (NN'25)	4.65/38.40	65.05/77.85	59.84/82.14	6.36/6.40	31.34/69.32	86.65/92.35	42.32/61.08
ERF (TASE'25)	51.00/83.82	71.26/80.09	53.15/72.75	70.53/87.41	48.28/59.97	82.18/94.37	62.73/79.74
UDASS (TPAMI'25)	32.84/57.54	74.10/81.07	**62.61**/91.43	**74.19**/91.69	54.22/63.89	**87.22**/97.08	64.20/80.45
MEBS (TGRS'25)	52.53/85.73	71.47/77.44	43.03/95.56	69.84/**92.80**	48.50/55.72	85.28/95.84	61.77/83.85
MCDA (TGRS'25)	45.04/57.88	73.23/80.10	55.37/69.31	70.57/90.76	50.33/59.94	83.56/96.11	63.02/75.68
DistilledDA [37] (WACV'26)	50.88/89.81	71.28/78.11	50.98/74.53	70.97/86.44	50.36/63.75	81.83/94.84	62.72/81.25
UCDS [38] (TIP'26)	48.81/88.44	72.30/79.65	60.28/79.82	72.58/89.36	49.78/61.12	84.66/96.54	64.74/82.49
PMPCA (ours)	**58.45/93.39**	**74.23**/82.01	47.96/**95.75**	72.85/86.94	**56.15**/65.35	85.28/**97.22**	**65.82/86.78**

classification accuracy for each category. They are defined as

$$\text{IoU} = \frac{TP}{TP + FP + FN}, \quad \text{ACC} = \frac{TP}{TP + FN} \tag{34}$$

where *TP*, *FP*, and *FN* denote the numbers of true-positive, false-positive, and false-negative instance pixels, respectively. The mean IoU (mIoU) is the average IoU over all categories, and the mean ACC (mACC) is the average ACC over all categories. Both IoU and ACC are bounded within the range $[0, 1]$, and higher values indicate better segmentation performance. For consistency and readability, all results reported in the tables are converted into percentages (%).

2) Comparison Methods: PMPCA is compared with 12 state-of-the-art UDA methods for semantic segmentation, including MemoryNet [24] (TGRS'23), UDARS [32] (RS'23), ERF [35] (TASE'25), PiPa [33] (ACM MM'23), SePico [10] (TPAMI'23), ST-DASeg [34] (JAG'24), PPPC [21] (NN'25), UDASS [3] (TPAMI'25), MEBS [36] (TGRS'25), MCDA [5] (TGRS'25), DistilledDA [37] (WACV'26), and UCDS [38] (TIP'26). These methods are trained and evaluated using their default configurations.

C. Results on the ISPRS Dataset

Tables I–IV present the semantic segmentation results of different UDA methods on the ISPRS dataset. Overall, PMPCA achieves the best mIoU across the four UDA settings, indicating its stronger adaptation capability under both cross-region and cross-spectrum domain shifts. Specifically, for

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TABLE IV

CROSS-DOMAIN SEMANTIC SEGMENTATION PERFORMANCE OF DIFFERENT UDA METHODS ON THE ISPRS DATASET UNDER THE VAI → POT-RGB SETTING. RESULTS ARE REPORTED AS IOU/ACC. BEST VALUE IN BOLD AND SECOND BEST VALUE IN UNDERLINE

Method	clutter	impervious surface	car	tree	low vegetation	building	avg
MemoryNet (TGRS'23)	7.21/8.11	58.51/75.10	52.53/59.19	38.56/59.99	49.52/61.46	54.31/84.10	43.44/57.99
UDARS (RS'23)	6.46/6.59	65.67/72.15	56.83/94.65	55.57/70.28	66.51/85.06	73.02/96.12	54.01/70.81
PiPa (ACM MM'23)	7.63/7.69	67.58/77.98	52.32/**95.53**	59.90/71.03	69.37/86.78	76.30/96.01	55.52/72.50
SePico (TPAMI'23)	**15.90/27.45**	63.65/82.48	61.43/92.83	46.63/59.42	63.46/72.57	74.88/95.88	54.32/71.77
ST-DASeg (JAG'24)	5.86/7.04	54.98/67.18	54.64/92.67	50.67/58.79	59.93/83.42	70.73/88.77	49.47/66.31
PPPC (NN'25)	5.76/5.87	62.34/79.01	68.44/84.60	29.68/**72.52**	19.58/22.84	71.52/93.27	42.88/59.68
ERF (TASE'25)	6.15/6.19	65.60/80.07	63.39/88.78	52.68/59.66	64.97/82.06	67.65/**97.01**	53.41/68.96
UDASS (TPAMI'25)	5.36/5.41	69.42/**85.23**	63.52/89.23	54.95/58.89	68.27/**89.15**	80.04/95.98	56.93/70.65
MEBS (TGRS'25)	10.59/10.95	67.29/81.41	56.24/88.92	52.34/57.70	68.31/87.67	74.62/96.27	54.90/70.49
MCDA (TGRS'25)	1.70/1.70	68.80/84.95	59.75/87.98	57.27/65.00	67.56/85.22	78.29/96.88	55.56/70.29
DistilledDA [37] (WACV'26)	10.68/10.96	69.76/85.03	58.82/87.59	51.46/57.02	67.79/87.45	77.90/96.08	56.07/70.69
UCDS [38] (TIP'26)	10.16/10.31	68.29/81.93	59.99/93.78	**61.79**/70.69	69.07/86.39	78.16/93.98	57.91/**72.85**
PMPCA (ours)	8.78/8.93	**70.11**/84.38	**69.03**/92.31	52.22/61.46	**69.96**/88.36	**80.42**/95.92	**58.42**/71.89

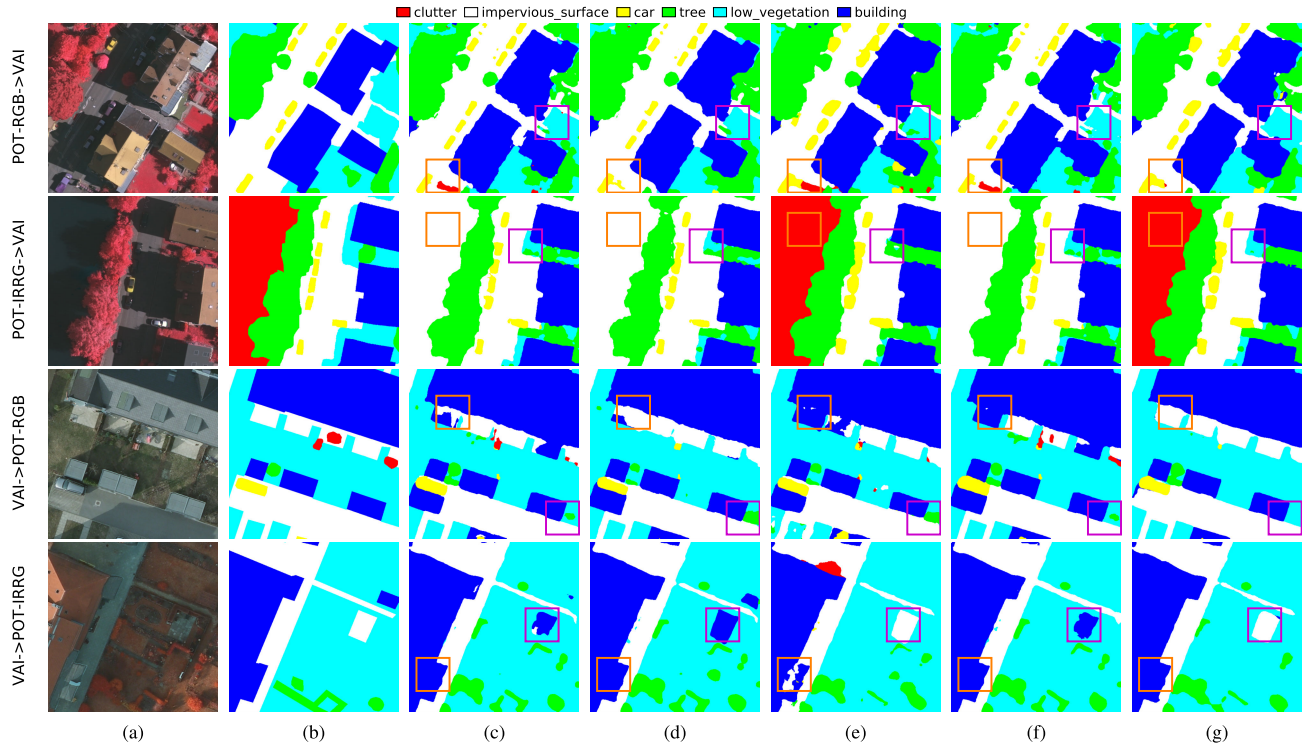


Fig. 7. Cross-domain semantic segmentation results of different methods on the ISPRS dataset. (a) Input RSI, (b) ground truth, (c) UDASS [3], (d) MEBS [36], (e) MCDA [5], (f) UCDS [38], and (g) PMPCA (ours).

LV, PMPCA obtains better IoU than UDASS and MEBS in POT-IRRG → VAI (58.43 versus 57.71 and 51.09), and also outperforms UDASS in POT-RGB → VAI (56.15 versus 54.22), suggesting its ability to handle categories with similar appearances and complex boundaries. In the VAI-to-POT settings, PMPCA performs favorably on IS and buildings, indicating that it can reduce confusion among large man-made structures. In addition, PMPCA shows strong performance on the small-scale car, outperforming MCDA and UDASS in VAI → POT-IRRG (86.03 versus 83.50 and 80.59). These results demonstrate that PMPCA excels at handling visually confusing and structurally diverse categories.

Fig. 7 visualizes the segmentation results on the ISPRS dataset under four UDA settings. First, existing UDA methods often suffer from category confusion among visually similar classes, such as IS versus clutter/background (POT-IRRG → VAI and POT-RGB → VAI), IS versus buildings (VAI → POT-RGB and VAI → POT-IRRG), and LV versus trees (POT-RGB → VAI). In contrast, PMPCA better distinguishes these categories and produces accurate semantic prediction. Second, existing UDA methods typically generate fragmented regions for buildings and vegetation areas (POT-IRRG → VAI and VAI → POT-RGB). In contrast, PMPCA preserves more complete structures and improves the spatial consistency of these regions. These results show the superiority of PMPCA on the ISPRS dataset.

D. Results on the LoveDA Dataset

Tables V and VI report the semantic segmentation performance of different UDA methods on the LoveDA dataset. PMPCA consistently achieves the best mIoU across two UDA settings, indicating its superior generalization ability under cross-region domain shift. Meanwhile, PMPCA shows

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TABLE V

CROSS-DOMAIN SEMANTIC SEGMENTATION PERFORMANCE OF DIFFERENT UDA METHODS ON THE LOVE DATASET UNDER THE RUR → URB SETTING. RESULTS ARE REPORTED AS IOU/ACC. BEST VALUE IN BOLD AND SECOND BEST VALUE IN UNDERLINE

Method	background	building	road	water	barren	forest	agriculture	avg
MemoryNet (TGRS'23)	36.05/63.83	49.87/62.29	48.95/58.18	67.86/74.63	38.33/58.25	39.48/62.24	39.24/47.43	45.68/60.98
UDARS (RS'23)	34.07/46.44	49.38/65.10	52.63/69.16	69.40/85.30	47.41/73.84	49.04/67.70	51.22/**69.86**	50.45/68.20
SePico (TPAMI'23)	38.84/64.43	**59.86/79.19**	57.47/**79.49**	58.13/67.39	46.38/58.96	49.86/60.97	51.33/59.72	51.69/67.16
PPPC (NN'25)	34.12/**72.76**	49.12/59.37	43.91/52.62	64.10/78.28	41.26/60.81	39.35/46.63	35.25/37.49	43.87/58.28
ERF (TASE'25)	39.33/66.90	55.60/71.86	56.21/67.71	67.46/83.24	39.46/44.82	49.40/59.28	49.12/59.34	50.94/64.73
UDASS (TPAMI'25)	33.45/44.85	54.05/75.01	53.89/64.84	66.68/84.86	46.70/76.79	50.50/**79.52**	55.90/67.80	51.60/70.52
MEBS (TGRS'25)	38.37/65.28	56.89/73.16	53.07/66.33	66.87/83.62	48.43/62.18	51.87/73.15	48.47/51.57	52.00/67.90
MCDA (TGRS'25)	38.44/72.43	58.22/75.00	52.83/58.55	70.07/84.74	46.83/58.39	47.08/60.41	43.90/46.25	51.05/65.11
DistilledDA [37] (WACV'26)	33.67/45.13	55.40/75.03	54.28/71.00	68.49/84.64	45.63/77.65	47.79/71.78	**56.81**/69.63	51.72/**70.69**
UCDS [38] (TIP'26)	37.83/59.17	55.87/70.10	55.85/68.02	71.65/85.24	46.57/**79.93**	43.26/53.55	54.07/65.34	52.16/68.76
PMPCA (ours)	**39.96**/64.60	51.96/63.98	**57.92**/71.91	**74.01**/83.23	45.04/52.62	**52.10**/67.93	50.49/63.77	**53.07**/66.86

TABLE VI

CROSS-DOMAIN SEMANTIC SEGMENTATION PERFORMANCE OF DIFFERENT UDA METHODS ON THE LOVE DATASET UNDER THE URB → RUR SETTING. RESULTS ARE REPORTED AS IOU/ACC. BEST VALUE IN BOLD AND SECOND BEST VALUE IN UNDERLINE

Method	background	building	road	water	barren	forest	agriculture	avg
MemoryNet (TGRS'23)	43.49/63.06	28.37/60.50	31.77/37.99	44.54/55.94	7.80/19.02	15.67/41.32	42.60/49.48	30.61/46.76
UDARS (RS'23)	52.56/67.19	15.88/**91.40**	37.25/**67.94**	57.88/**92.67**	**23.21/30.54**	31.92/47.38	20.33/22.20	34.15/**59.90**
PiPa (ACM MM'23)	**57.61/93.08**	43.44/58.39	43.93/48.56	**69.71**/81.90	17.69/21.58	2.43/2.47	45.82/47.25	40.09/50.46
SePico (TPAMI'23)	55.41/83.18	45.92/54.36	32.82/47.51	59.29/82.68	15.92/20.79	30.73/45.91	45.62/48.65	40.82/54.73
PPPC (NN'25)	41.37/61.25	32.34/68.90	33.47/42.98	42.50/80.01	12.73/26.31	21.46/56.54	27.29/27.81	30.17/51.97
ERF (TASE'25)	48.67/84.42	48.93/66.37	43.89/50.89	69.25/84.79	15.77/18.27	31.38/52.16	20.75/21.20	39.81/54.01
UDASS (TPAMI'25)	49.88/86.30	47.37/65.01	46.24/50.98	68.67/85.16	12.75/13.88	**38.47**/56.41	23.05/23.40	40.92/54.45
MCDA (TGRS'25)	50.26/75.13	52.29/62.36	40.77/47.93	60.29/77.28	13.16/17.72	28.54/48.03	49.71/56.95	42.15/55.06
DistilledDA [37] (WACV'26)	54.94/91.86	39.86/71.93	39.80/49.65	68.45/73.93	6.98/7.38	20.21/23.84	36.06/37.04	38.04/50.81
UCDS [38] (TIP'26)	57.34/85.80	53.97/69.65	31.70/32.36	65.99/72.31	15.84/29.97	18.37/19.54	**56.41/62.33**	42.80/53.14
PMPCA (ours)	51.54/83.93	**57.25**/73.18	**47.23**/51.95	66.36/85.79	19.56/23.76	32.92/51.63	32.43/33.50	**43.90**/57.68

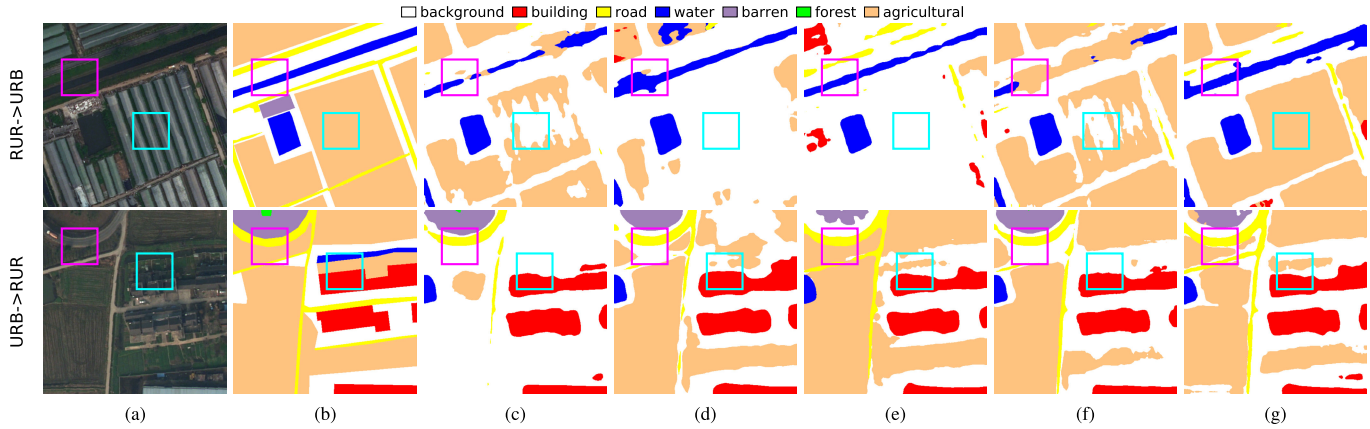


Fig. 8. Cross-domain semantic segmentation results of different methods on the LoveDA dataset. (a) Input RSI, (b) ground truth, (c) UDASS [3], (d) MEBS [36], (e) MCDA [5], (f) UCDS [38], and (g) PMPCA (ours).

advantages for structurally continuous and visually confusing classes. In RUR → URB, PMPCA achieves better results for road than UCDS (57.92 versus 55.85), suggesting that it better preserves spatial continuity in complex urban scenes. In URB → RUR, PMPCA obtains the best results for building and road, outperforming MCDA on building (57.25 versus 52.29) and UDASS on road (47.23 versus 46.24). These results demonstrate that PMPCA can better distinguish man-made structures and preserve continuous road regions under challenging domain shifts.

Fig. 8 presents the segmentation results on the LoveDA dataset under two UDA settings. Existing UDA methods identify part of the road or farmland, and incorrectly assign farmland interiors to the background, leading to incomplete structures (RUR → URB). PMPCA better preserves the continuity of farmland while accurately distinguishing nearby water and road regions. In the URB → RUR, narrow roads, buildings, and the background are easily confused. PMPCA produces more coherent predictions and clearer distinctions among these categories. These results show that PMPCA is effective in maintaining semantic integrity and improving cross-domain segmentation on the LoveDA dataset.

E. Ablation Studies

Ablation experiments are conducted on the ISPRS dataset under the POT-IRRG → VAI setting to evaluate the effects of

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TABLE VII

ABLATION STUDIES OF PMPCA ON THE POT-IRRG → VAI SETTING. UCA: UNCERTAINTY-GUIDED CURRICULUM ADAPTATION STRATEGY; PMPL: PROBABILISTIC MULTIPLE PROTOTYPES LEARNING MODULE; PWC: PRIOR-WEIGHTED CONTRASTIVE CONSTRAINT; WEIGHTING: CATEGORY-FREQUENCY WEIGHTING IN PWC. BEST VALUE IN BOLD AND SECOND BEST VALUE IN UNDERLINE

Method	Weighting	mIoU	mACC
Baseline	–	64.02	81.87
+ UCA	–	65.89	85.81
+ PMPL	–	66.76	85.09
+ UCA + PMPL	–	68.13	**87.44**
+ PMPL + PWC	–	67.18	85.99
+ UCA + PMPL + PWC	w/o	<u>68.15</u>	86.57
+ UCA + PMPL + PWC	w/	**68.83**	<u>87.15</u>

the PMPL module, PWC constraint, and UCA strategy. The results are reported in Table VII.

1) Baseline: The baseline setup is described in Section III-A. It yields a relatively low mIoU/mACC of 64.02/81.87, indicating that simple ST is insufficient for effective UDA.

2) Effect of UCA Strategy: To improve ST, the UCA strategy performs progressive UDA from easy to hard scenarios. As shown in Table VII, introducing UCA improves the baseline performance (65.89/85.81 versus 64.02/81.87), indicating that scheduling target-domain RSIs according to scenario difficulty can reduce the influence of noisy pseudo-labels during early adaptation.

3) Effect of PMPL Module: The PMPL module represents category distributions in a fine-grained manner. As shown in Table VII, PMPL improves the baseline performance (66.76/85.09 versus 64.02/81.87), demonstrating its effectiveness in preserving category semantics across domains.

4) Effect of PWC Constraint: The PWC constraint performs structured alignment with unbiased priors across domains. Since PWC is defined on the prototype space, it is evaluated on top of the PMPL module. As shown in Table VII, combining PMPL with PWC further improves the performance over PMPL alone (67.18/85.99 versus 66.76/85.09), verifying its ability to refine the learned prototype space.

To verify the prior weighting in the PWC constraint, we compare its weighted and unweighted versions. The results in Table VII show that the weighted version achieves better performance (68.83/87.15 versus 68.15/86.57), indicating that domain-specific category-frequency weighting alleviates category-prior mismatch and further improves cross-domain structured alignment.

F. Hyperparameter Configurations of Prototypes

Fig. 9 illustrates the semantic segmentation performance on the ISPRS dataset under the POT-IRRG → VAI setting with different numbers and dimensions of prototypes. The results show that the mIoU first increases and then decreases as either the number or the dimension of prototypes increases.

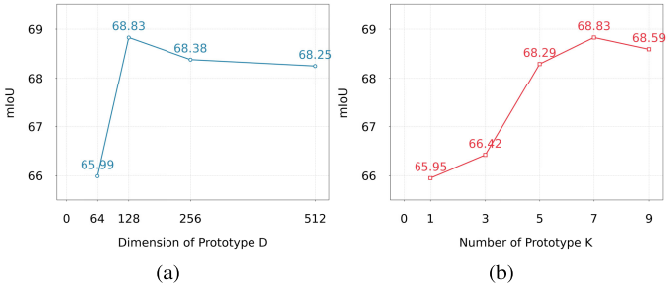


Fig. 9. Effect of number and dimension of prototypes on cross-domain segmentation performance. (a) Number of prototypes. (b) Dimension of prototypes.

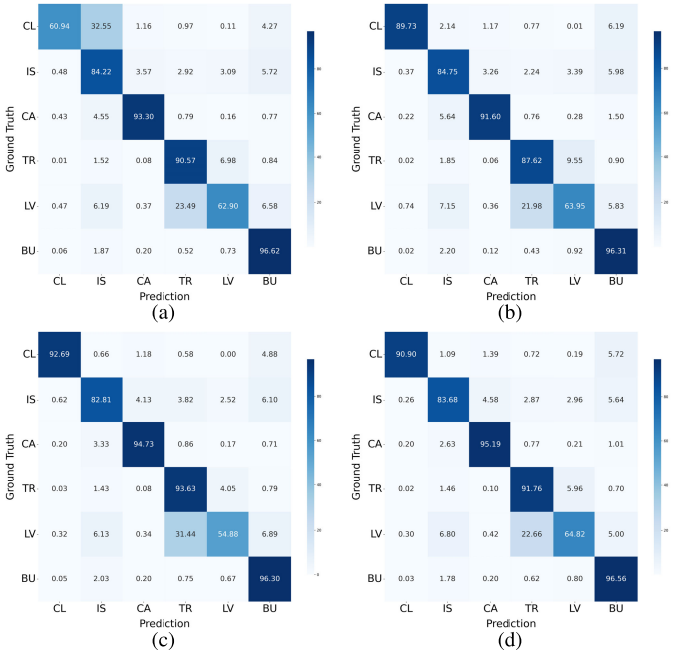


Fig. 10. Confusion matrix comparison of different types of prototypes. (a) Centroid-based single prototype. (b) Centroid-based multiple prototypes. (c) Probabilistic single prototype. (d) Probabilistic multiple prototypes. CL (clutter), IS, CA (car), TR (tree), LV, and BU (building).

When the number of prototypes is too small, or the prototype dimension is too low, their semantic capacity is insufficient. In contrast, when the number of prototypes is too large, or the prototype dimension is excessively high, the prototypes become overly dispersed. In both cases, the representation ability of prototypes is degraded, making it difficult for the model to effectively capture complex category distributions across domains. Therefore, the optimal number and dimension of prototypes are 7 and 128, respectively.

G. Comparison of Different Types of Prototypes

Fig. 10 compares the semantic segmentation performance (confusion matrix) of different types of prototypes on the ISPRS dataset, using the POT→VAI UDA setting. First, single-centroid and multicentroid prototypes represent each category distribution with a single centroid and multiple centroids, respectively. In contrast, the multicentroid prototype more effectively captures intracategory variability, achieving

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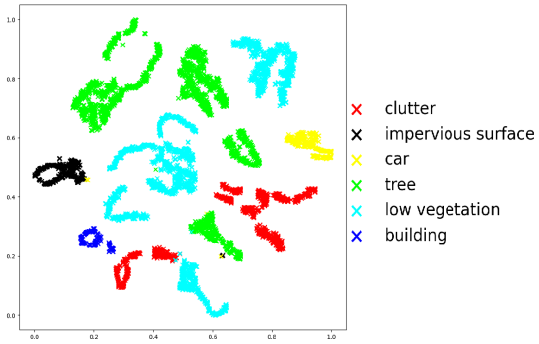


Fig. 11. T-SNE visualization of prototypes.

an IoU of 66.63 and ACC of 85.66, with improvements of 2.71% and 4.25% over the single-centroid prototype, respectively. Second, single-centroid and single-distribution prototypes represent each category with a single centroid and a single probability distribution, respectively. In contrast, the single-distribution prototype accounts for both the central location and dispersion shape of the category, resulting in an IoU of 65.44 and an ACC of 85.84, improving by 1.52% and 4.43% over the single-centroid prototype, respectively. This indicates that category distributions are anisotropic. Finally, the multicentroid prototype represents each category with multiple centroids and multiple probability distributions. In contrast, the multidistribution prototype not only preserves intracategory variability but also more accurately captures the similarity between categories, resulting in clearer decision boundaries. Therefore, the multidistribution prototype achieves the highest IoU of 68.83, improving by 2.20% over the multicentroid prototype.

H. Visualization Analysis of Prototypes

To verify the effectiveness of the PMPL module, the prototypes are visualized using t-SNE, as shown in Fig. 11. The experimental results on the ISPRS dataset, using the POT-IRRG → VAI setting, show that: first, the prototypes of different categories are obviously separated from each other in the feature space. However, the prototypes of the same category remain relatively clustered; although the distance between the partial prototypes is large, they are still located in the cluster of the corresponding category. This indicates that the PMPL module can effectively represent the category distribution across domains. On the one hand, each prototype represents different patterns within the same category, enabling the model to distinguish the diverse features within the category. On the other hand, probabilistic parameters allow different patterns to have their own shapes, which makes the decision boundary of different categories clearer.

I. Visualization Analysis of UCA

To verify the effect of the UCA strategy, Fig. 12 presents representative examples of easy and hard scenarios in the target domain. The results show that easy scenarios tend to generate more reliable pseudo-labels with clearer semantic structures, whereas hard scenarios are more likely to involve complex category co-occurrence or spatial layouts, resulting in noisier pseudo-labels. By adopting the easy-to-hard scenario adaptation in UCA, the negative influence of noisy pseudo-labels can be reduced during training.

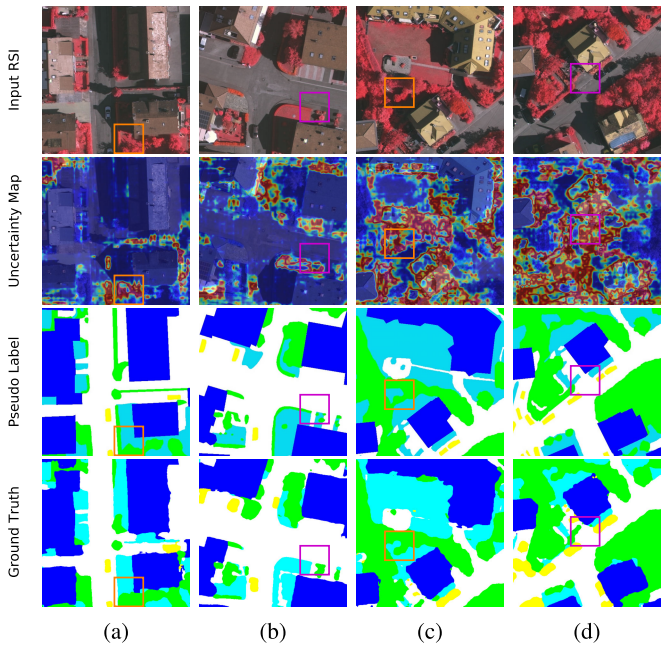


Fig. 12. Visualization analysis of the UCA strategy. (a) and (b) Easy scenarios. (c) and (d) Hard scenarios.

TABLE VIII
COMPARISON BETWEEN GLOBAL UCA AND GLOBAL-TO-LOCAL UCA VARIANT. BEST VALUE IN BOLD AND SECOND BEST VALUE IN UNDERLINE

Strategy	mIoU	mACC
Global	**68.83**	<u>87.15</u>
Global-to-Local	<u>68.49</u>	**87.30**

J. Comparison Between Global UCA and Global-to-Local UCA Variant

Easy and hard regions may coexist within a single-target domain RSI, reflecting intraimage heterogeneity. To investigate this issue, we design a global-to-local UCA variant. Specifically, under the global UCA, target domain RSIs are first divided into easy and hard subsets according to global image-level uncertainty. Then, local region-level uncertainty is estimated within each selected RSI. During the first half of training, local regions whose uncertainty exceeds the image-level mean are excluded from the target domain constraint, and they are gradually introduced during the second half of training. The results are reported in Table VIII. The global-to-local UCA variant slightly improves mACC but decreases mIoU. This phenomenon can be attributed to two factors. First, the global UCA adopts the Q1 quartile, meaning that only the top 25% of target domain RSIs are introduced. These selected RSIs are generally simple, and thus their local ambiguity is relatively limited. Second, the pseudo-label confidence threshold is set to 0.968, which is strict to filter out

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TABLE IX
SENSITIVITY ANALYSIS OF KEY HYPERPARAMETERS. BEST VALUE IN BOLD AND SECOND BEST VALUE IN UNDERLINE

Parameter	Setting	mIoU	mACC
Balance factor λ	0.5	68.49	**87.30**
	1.0	**68.83**	87.15
	1.5	68.79	85.74
Temperature τ	0.05	68.47	86.13
	0.10	**68.83**	87.15
	0.20	68.64	**87.71**
Quantile γ	0.25 (Q1)	**68.83**	**87.15**
	0.50 (Q2)	67.47	86.67
	0.75 (Q3)	66.21	81.88

many uncertain pixels. In addition, although local uncertainty estimation may suppress noisy regions, it may also affect the learning of informative hard regions.

K. Sensitivity Analysis of Hyper-Parameters

1) Sensitivity Analysis of Temperature: The temperature parameter τ in the PWC constraint controls the sharpness of the contrastive distribution. To evaluate its influence, we vary τ among {0.05, 0.10, 0.20}. As reported in Table IX, PMPCA achieves the best mIoU when $\tau = 0.10$. When τ is smaller, the contrastive distribution becomes overly sharp, causing the model to focus on fewer similar instance-prototype pairs and become more sensitive to noisy pseudo-labels. When τ is larger, the contrastive distribution becomes overly smooth, which weakens the discriminative ability of the learned prototypes.

2) Sensitivity Analysis of Balance Factor: The balance factor λ controls the relative importance of mixed-domain consistency and the PWC constraints with respect to the source domain supervised constraint. To analyze its effect, we vary λ among {0.5, 1.0, 1.5}. As shown in Table IX, PMPCA obtains the best overall performance when λ is set to 1.0. When λ is smaller, mixed-domain consistency and the PWC constraints are weakened, hindering cross-domain alignment and prototype discrimination. When λ is larger, mixed-domain consistency and the PWC constraints become overly emphasized, which may over-regularize domain alignment.

3) Sensitivity Analysis of Quantile: The quantile γ in the UCA strategy controls the easy-to-hard scenario ratio of the target domain introduced during curriculum adaptation. To study its effect, we evaluate three quartile-based settings, namely 0.25, 0.50, and 0.75, corresponding to Q1, Q2, and Q3, respectively. As shown in Table IX, PMPCA achieves the best performance when γ is set to 0.25. When γ becomes larger, more difficult target domain scenarios are introduced, which increases the risk of noisy pseudo-label propagation.

L. Efficiency of PMPCA

To evaluate the efficiency of PMPCA, we compare it with two representative UDA methods, UDASS [3] and MCDA [5], under an input RSI size of 512 × 512 × 3 and a training batch size of 4. Table X reports the number of parameters (Params), floating-point operations (FLOPs), training GPU memory (Train Mem), training time per batch (Train Time), testing GPU memory (Test Mem), and testing time (Test Time) in frames per second (fps). As shown in Table X, PMPCA maintains a comparable model size to UDASS and MCDA, indicating that its performance gain does not rely on increased model capacity. Meanwhile, PMPCA requires fewer FLOPs and achieves higher testing fps than UDASS and MCDA. Although PMPCA incurs additional training overhead due to the PMPL module and the PWC constraint, they are only introduced during training to improve cross-domain adaptation. Therefore, PMPCA achieves a favorable tradeoff between adaptation performance and practical applicability.

TABLE X
EFFICIENCY COMPARISON BETWEEN PMPCA AND RECENT BASELINES UNDER THE INPUT RSI SIZE OF 512 × 512 × 3 AND A TRAINING BATCH SIZE OF 4. BEST VALUE IN BOLD AND SECOND BEST VALUE IN UNDERLINE

Method	Params (M)	GFLOPs	Train Mem (GB)	Train Time (s/batch)	Test Mem (GB)	Test Time (fps)
UDASS [3]	85.68	302.60	**11.02**	**1.74**	3.42	7.31
MCDA [5]	86.50	144.94	14.10	2.38	3.67	6.83
PMPCA (ours)	**85.61**	**126.63**	13.08	3.37	**2.68**	**15.22**

V. LIMITATIONS AND FUTURE WORK

Despite the remarkable performance of PMPCA, several limitations remain. First, although probabilistic multiple prototypes improve the fine-grained representation of category distributions, their construction and assignment introduce additional training overhead, increasing the iteration time of each training batch. Second, the curriculum adaptation estimates difficulty at the image level, providing an effective data scheduling strategy but failing to fully capture fine-grained heterogeneity within a target domain RSI.

These limitations suggest several future directions. First, more efficient prototype construction and assignment strategies can be explored to reduce the computational cost during training while preserving probabilistic representation. Second, region-aware curriculum adaptation can be further expanded by region priors from the Segment Anything Model (SAM) [41], since regions are often more semantically homogeneous. In addition, broader validation under more challenging domain shifts, including cross-temporal adaptation and cross-quality adaptation, would be valuable for assessing the robustness of PMPCA.

VI. CONCLUSION

This work focuses on UDA for RSI semantic segmentation, aiming to address the challenges of complex category distributions within domains and mismatched category priors across domains. The PMPL module accurately represents category distributions by constructing probabilistic multiple prototypes, which preserve the semantic integrity of each category across domains. The PWC constraint adaptively calibrates domain alignment progress by category frequency weighting, which mitigates category prior bias across domains. In addition, the UCA strategy is designed to achieve progressive UDA, transitioning from easy to hard scenarios. This enhances the efficiency of the self-training process. Extensive experiments

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demonstrate that the proposed method achieves state-of-the-art performance and exhibits remarkable generalization across cross-spectrum bands and cross-space regions.

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